Lesson plan · for the teacher

Standard deviation and outliers

Data science · Level 2 · Lesson 2

Time
40 min

Materials
Board and chalk or markers; Paper and pencils; A phone or tablet with Darsi, for the lab

Learning goals

- Read a standard deviation.
- Flag outliers with the IQR rule.
- Check outliers before acting.

Activity: Spread detectives

1. Warm-up (5 min): Show two classes with a mean of 12, one tight and one spread out. Ask what a teacher would do differently.
2. Teach (8 min): Show distances from the mean as arrows; then find quartiles and draw the fences.
3. Find (15 min): Groups get a list of journey times with two odd values. They work out the fences and flag outliers.
4. Investigate (7 min): For each outlier, groups decide what could explain it and how to check.
5. Wrap-up (5 min): Ask two pupils to say the finding in one sentence with a number in it. Then give the exit ticket.

Discussion

- When does a large SD matter to a teacher?
- Why not delete outliers?
- Which outlier surprised you?

Exit ticket

- Q1 is 20 and Q3 is 30. Write the two fences.

Try it in class tomorrow
Take last term's marks without names, find the mean and the standard deviation, and look for outliers. Discuss what each outlier might mean.

Worksheet · for pupils

Standard deviation and outliers

Data science · Level 2 · Lesson 2

Name _____ Class _____ Date _____

1 **Class C has a standard deviation of 2, class D of 6. Which has more mixed marks?**

◯ Class C ◯ Class D ◯ They are the same

2 **An outlier should always be deleted so it doesn't spoil the average.**

◯ True ◯ False

3 **Print every mark that is more than two standard deviations above the mean, as a whole number.**

```
from ulab import numpy as np
marks = np.array([11, 12, 12, 13, 12, 30])
m = np.mean(marks)
s = np.std(marks)
print(m)
```

Write your program:

4 **Class A's marks are 11, 12, 12, 13. Class B's are 5, 9, 15, 19. Both have a mean of 12. Which has the larger standard deviation?**

◯ Class B ◯ Class A ◯ They are equal

5 **Complete the rule.**

Q1 is 10 and Q3 is 14, so the IQR is _____ . A mark above _____ is flagged as an outlier.

Word bank: 4 · 6 · 18 · 20

✎ **Think of a pupil whose numbers stand out. What would the standard deviation have hidden, and what does the outlier tell you?**